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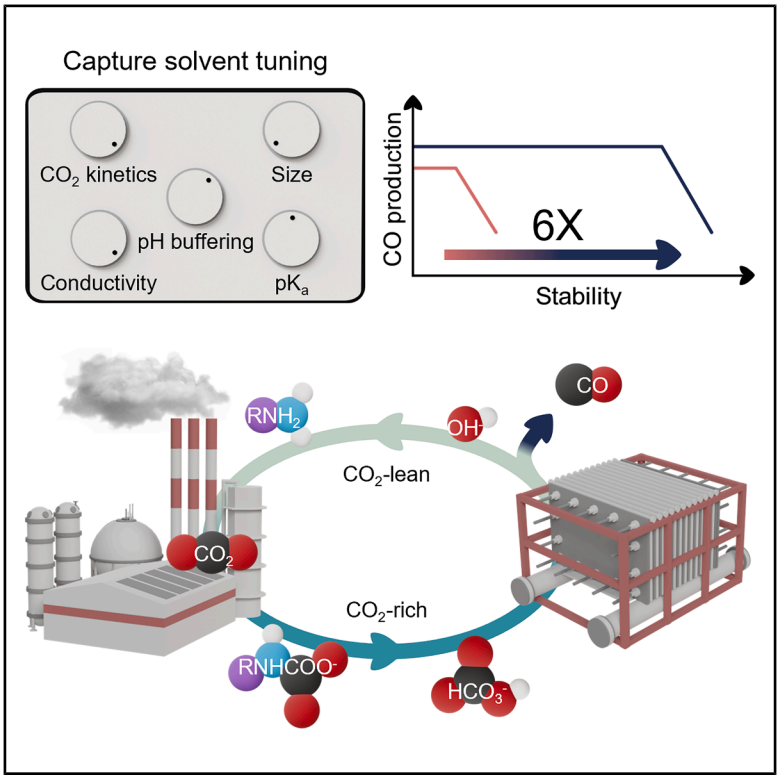
Solvent tuning regulates proton flux to extend stability in reactive CO_2 capture and electrolysis

Graphical abstract



Highlights

- Solvent tuning regulates the proton flux to enhance CO selectivity and stability
- Balancing cathode pH extends CO_2 conversion stability to over 250 h
- Integrated CO_2 capture and conversion is demonstrated in a closed-loop system
- Higher CO_2 conversion rates improve *in situ* capture solvent regeneration

Authors

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In brief

Integrated CO_2 capture and electrochemical conversion offer energy-efficient chemical production, though commercial viability is currently limited by low electrolysis stability. This work demonstrates that solvent tuning effectively regulates interfacial proton flux, significantly extending cell durability in a closed-loop system to advance practical carbon utilization.

Xiao et al., 2026, Chem Catalysis *6*, 101694
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https://doi.org/10.1016/j.checat.2026.101694

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Please cite this article in press as: Xiao et al., Solvent tuning regulates proton flux to extend stability in reactive CO_2 capture and electrolysis, Chem Catalysis (2026), https://doi.org/10.1016/j.checat.2026.101694

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Solvent tuning regulates proton flux to extend stability in reactive CO_2 capture and electrolysis

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https://doi.org/10.1016/j.checat.2026.101694

THE BIGGER PICTURE Capturing and electrochemically converting CO_2 into value-added chemicals, such as CO, supports the transition to a net-zero emissions economy. Reactive capture of CO_2 (RCC) is an integrated system that directly feeds the reactor with CO_2-rich post-capture solutions, simultaneously producing high-purity gas products and regenerating the capture solvent. However, current RCC systems suffer from low electrolysis stability, a critical performance hurdle for industrialization. In this work, we analyze degradation pathways and identify that performance loss is dominated by processes localized at the catalytic interface, where high proton flux accelerates catalyst decomposition. We demonstrate that solvent tuning complements traditional catalyst innovations by offering a distinct set of tunable parameters that benefit the entire system. By regulating the cathode's local pH through capture solvent tuning and operating parameter optimization, we extended stability to over 75 h at 100 mA cm^{-2} and over 250 h at 50 mA cm^{-2}. We demonstrate the feasibility of this approach by generating a high-purity product stream in a closed-loop system. This solvent-tuning strategy provides a framework for stabilizing a wide range of electrochemical technologies governed by complex pH dynamics.

SUMMARY

Reactive capture of CO_2 (RCC) integrates capture and electrochemical conversion processes, offering energy-efficient CO production. RCC systems employing amino-acid-salt-based capture solvents, such as potassium glycinate (K-GLY), exhibit rapid CO_2 sorption kinetics and high CO selectivity yet are stable for fewer than 30 h. Lowering K-GLY concentrations unexpectedly reduced lifetime—an effect we attribute to reduced pH buffering and proton flux regulation. From these results, we identify a fundamental limitation of RCC systems: the protons that facilitate *in situ* CO_2 regeneration also accelerate catalyst degradation. By tuning the capture solvent composition and the operating parameters, we balance cathode pH and extend stability to over 75 h at 100 mA cm^{-2} and over 250 h at 50 mA cm^{-2}. In an integrated absorber-electrolyzer system, this approach generated a ~95 vol % product stream with ~5 vol % CO_2 and demonstrated an overall CO_2 utilization efficiency of 35%.

INTRODUCTION

The electrochemical reduction of CO_2 enables the conversion of captured CO_2—from point sources or the atmosphere—using renewable electricity, simultaneously mitigating anthropogenic CO_2 emissions and producing industrial chemical feedstocks such as CO.[1] However, regenerating captured CO_2 from solvents via traditional thermal-driven methods is energy intensive, accounting for ~20% of the total energy demand in conventional capture and conversion processes.[2] Furthermore, excess CO_2 in the product gas stream needs to be removed. Reactive capture of CO_2 (RCC) could enable energy-efficient and economical CO production by integrating the capture and conversion processes.[3] RCC uses CO_2-rich, post-capture solutions directly as reactor inputs, eliminating the CO_2 regeneration required by conventional gas-phase CO_2 electrolyzers and

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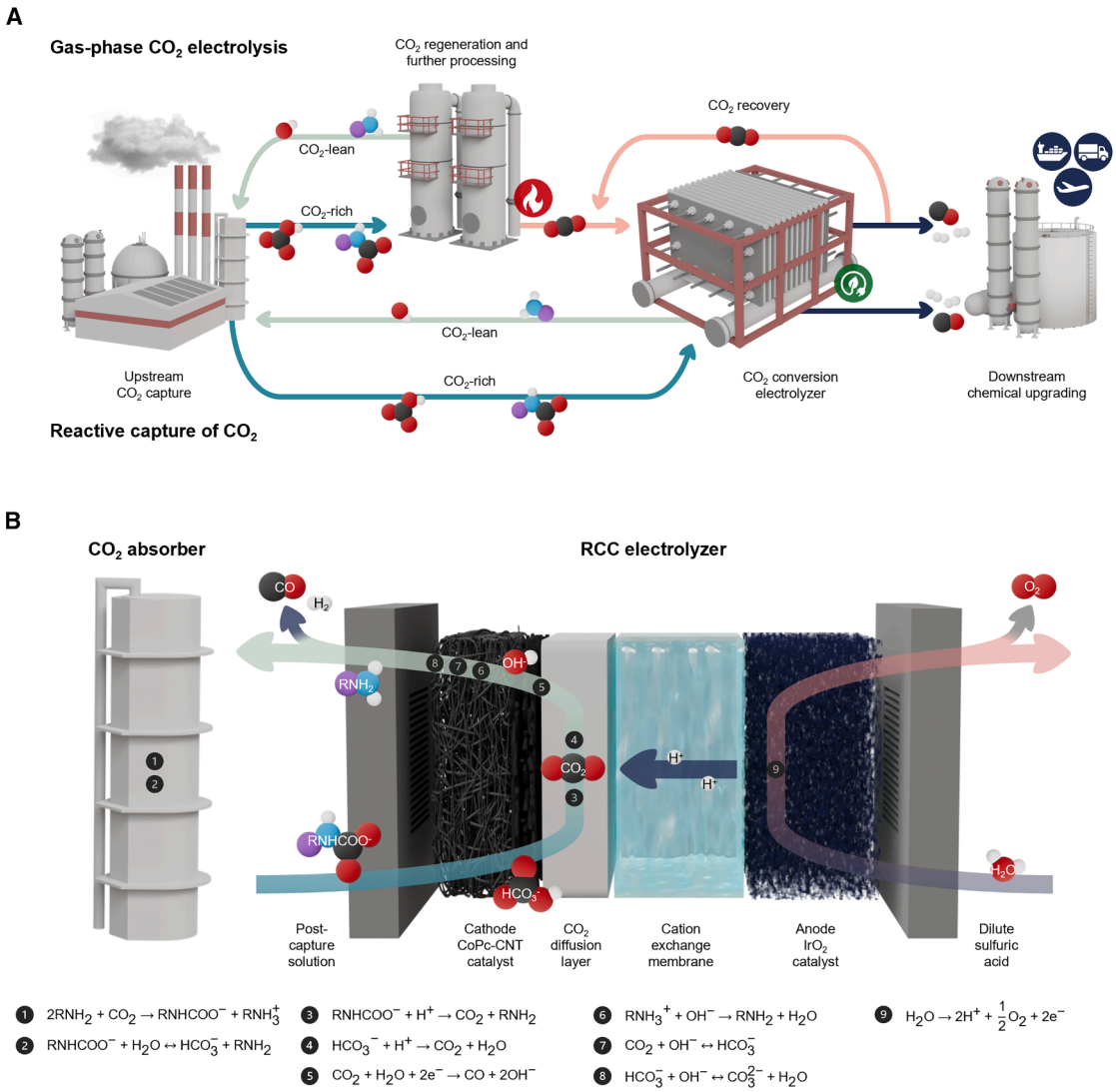

Figure 1. Reactive capture of CO_2 schematics

(A) Comparison of chemical upgrading via gas-phase CO_2 electrolysis (top) and reactive capture of CO_2 (RCC) (bottom), including major processes and the dominant chemical species involved in each process.

(B) CO_2 absorber and membrane electrode assembly for RCC, including electrolyzer components and chemical reactions.

minimizing excess CO_2 to reduce downstream product separation energy demand (Figure 1A).[4]

To industrialize and integrate with existing energy infrastructures, RCC systems need to demonstrate selective CO_2 conversion and robust stability.[5–7] Recent works have achieved >80% Faradaic efficiency (FE) toward CO at current densities >100 mA cm^{-2}.[8] However, these high conversion rates rely on CO_2-rich post-capture solutions, which are difficult to obtain using only inorganic hydroxide solvents that are not representative of industrial post-combustion capture processes.[9] Industrial CO_2 capture relies on amine-based solvents, such as alkanolamines and diamines, due to fast CO_2 absorption by their highly reactive amino functional groups.[10,11] Amino acid salts (AASs) (e.g., potassium glycinate [K-GLY]) are promising amine-based solvents that exhibit excellent CO_2 capture kinetics and high CO_2 loading capacity,[12] as well as favorable physiochemical properties over conventional amines, such as high ionic conductivity, low volatility, improved oxygen tolerance, and low toxicity.[13] Post-combustion CO_2 capture using an AAS solvent has been demonstrated on the pilot scale.[14] This combination of features makes AASs suitable capture solvents for RCC. Though selective CO production has been achieved in AAS-based RCC, current systems have demonstrated less than 30 h of continuous operation at a CO partial current density of ~30 mA cm^{-2} (Figure S1; Table S1).

In this work, we investigate the degradation mechanisms in AAS-based RCC using an established CO_2-to-CO molecular catalyst with limited catalyst-amine interactions. We identify the cathode's local pH as the primary contributing factor and extend electrolysis stability at 100 mA cm^{-2} by tuning the

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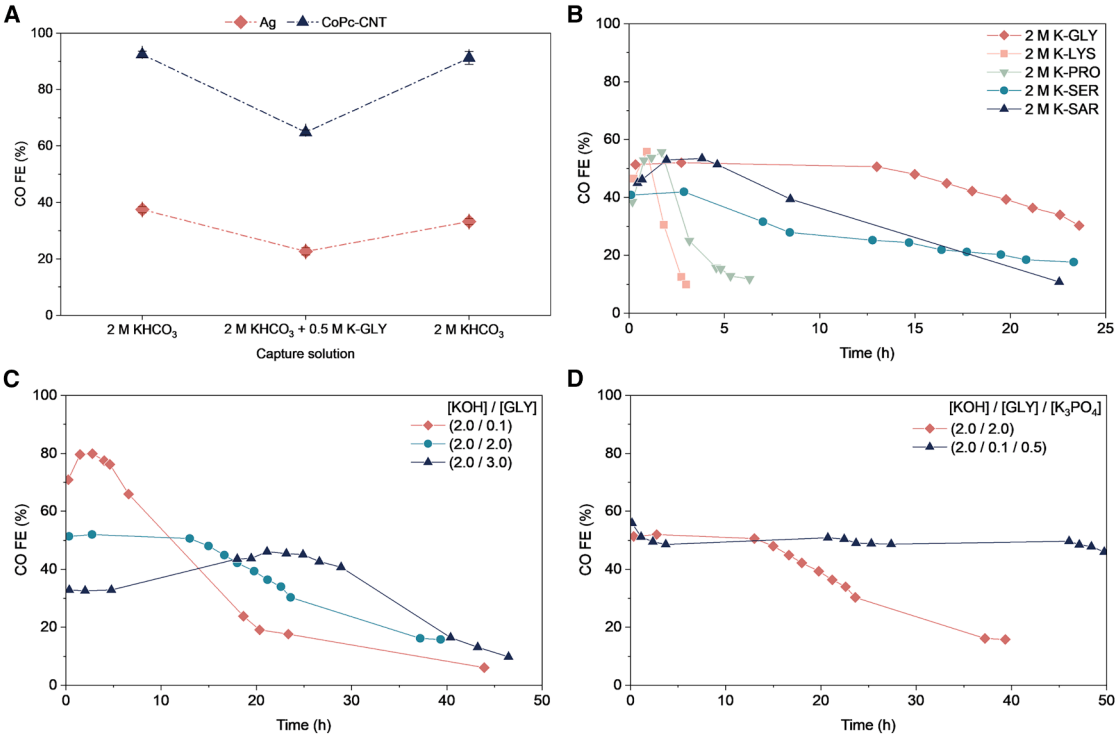

Figure 2. Optimization of the capture solvent composition

All experiments were conducted using a 1 cm^2 active area at an applied current density of 100 mA cm^{-2}.

(A) CO Faradaic efficiency (FE) when using 2 M $KHCO_3$ as the catholyte, then adding 0.5 M potassium glycinate (K-GLY) to the solution, then replacing the catholyte with fresh 2 M $KHCO_3$ for an Ag and a molecularly dispersed cobalt phthalocyanine (CoPc-CNT) catalyst. Error bars correspond to the standard deviation of at least three independent measurements.

(B) CO FE over time using different amino acid salt capture solvents: K-GLY, potassium lysinate (K-LYS), potassium prolinate (K-PRO), potassium serinate (K-SER), and potassium sarcosinate (K-SAR).

(C) CO FE over time using capture solvents consisting of 2.0 M KOH and glycine (GLY) concentrations ranging from 0.1 to 3.0 M.

(D) CO FE over time using a 2 M K-GLY capture solvent and one consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4.

capture solvent composition and the catholyte flow rate. We apply the capture solvent tuning strategy across a range of current densities, where differing hydroxide production rates result in varying inherent cathode alkalinity. In a closed-loop demonstration with the catholyte circulating between a CO_2 absorber, an RCC electrolyzer, and a gas-liquid separator, we evaluate the CO_2 capture and utilization efficiencies. The optimized system achieved over 75 h of stable operation with a CO partial current density of ~60 mA cm^{-2}, representing a 6-fold improvement over initial results. This work addresses the stability challenge in RCC systems—the largest barrier to commercialization—by presenting an effective and robust pH-tuning strategy.

RESULTS AND DISCUSSION

Capture solvent tuning

To study the stability of AAS-based RCC, we employed a CO-selective cathode and an iridium oxide anode in a membrane electrode assembly, separated by a cation exchange membrane that facilitated proton transport for *in situ* CO_2 regeneration via a pH swing, and a CO_2 diffusion layer that maintained local alkalinity at the cathode for efficient CO_2 conversion (Figure 1B).[15] CO_2-rich post-capture solutions were circulated on the cathode side while dilute sulfuric acid was used on the anode side to support proton production from the acidic oxygen evolution reaction. During electrolysis, the catholyte was sparged with 20 sccm of 15 vol % CO_2 (17 sccm nitrogen and 3 sccm CO_2), unless otherwise stated, to maintain CO_2 equilibrium and to avoid bulk reactant depletion.

Conventional silver (Ag)-based catalysts in potassium (bi)-carbonate ($KHCO_3$ and K_2CO_3) solutions have demonstrated up to 150 h of stability while producing CO with ~30% FE (Figure S2).[16] However, adding 0.5 M K-GLY into 2 M $KHCO_3$ lowered the CO FE from 37% to 23% (relative decrease of 38%) (Figure 2A). Upon replacing the K-GLY-containing catholyte with a fresh 2 M $KHCO_3$ solution, the CO FE recovered to 33% (89% of its original selectivity). We reasoned that this decrease in CO FE resulted primarily from strong Ag-amine interactions[17–19] rather than irreversible catalyst degradation. We posited that a molecularly dispersed catalyst, which interacts less with amines,[20] would lead to higher CO-selectivity retention in AAS-based RCC.[21] Among molecular catalysts, cobalt phthalocyanine supported on carbon nanotubes (CoPc-CNT) is a well-studied CO_2-to-CO benchmark catalyst that offers

high activity, chemical stability, and structural tunability, and its reliable and reproducible performance makes it a controlled baseline for studying RCC stability.[22–24] In bicarbonate-fed RCC electrolyzers, CoPc-CNT catalysts have demonstrated higher CO selectivity than Ag catalysts, attributed to their well-defined molecular active sites and favorable CO_2 binding and activation pathways.[25,26] With a CoPc-CNT catalyst, the CO FE increased to over 80% in 2 M $KHCO_3$ (Figure S3). Upon the addition of 0.5 M K-GLY, the CO FE decreased from 92% to 65% (relative decrease of 29%); however, the CO FE fully recovered to 91% (99% of its original selectivity) after electrolyte replacement with fresh 2 M $KHCO_3$. Comparing the relative CO FE decrease and recovery normalized to each catalyst's initial selectivity isolates K-GLY's impact, independent of baseline performance. The result suggests that CoPc-CNT offers better selectivity and greater amine tolerance than Ag-based catalysts. We therefore used CoPc-CNT for ensuing RCC studies. CO and hydrogen (H_2) were the only detectable electrolysis products.

Several degradation pathways have been proposed that could contribute to the loss of CO selectivity observed in amine-based RCC systems, including catalyst adsorption,[27] hindered CO_2 mass transport due to increased electrolyte viscosity,[28] and proton shuttling that promoted competitive hydrogen evolution.[29] We set out to study the degradation mechanisms in AAS-based RCC by first comparing its stability using different solvents. We prepared 2 M AAS solvents containing equimolar potassium hydroxide (KOH) and various amino acids: glycine (GLY; 75 g mol^{-1}), sarcosine (SAR; 89 g mol^{-1}), serine (SER; 105 g mol^{-1}), proline (PRO; 115 g mol^{-1}), and lysine (LYS; 146 g mol^{-1}) (Figures S4–S8). We evaluated CO FE over time for each AAS post-capture solution (Figure 2B; Table S2). The onset time of CO FE decline ranged from 1 to 13 h, with K-GLY exhibiting the highest performance.

Electrochemical impedance spectroscopy (EIS) analysis before and after prolonged electrolysis revealed that more stable systems exhibited a smaller increase in series and charge transfer resistances (R_s and R_{ct}, respectively) (Figures S9 and S10). R_s reflects the overall ohmic pathway and can increase due to the membrane or electrode material degradation, contact loss, or interfacial delamination. Failure mechanisms common in gas-phase CO_2 electrolyzers—such as salt precipitation or gas-diffusion layer flooding, which increases bulk resistance[30]—are less relevant under liquid-fed conditions with hydrophilic electrodes employed here. In contrast, an increase in R_{ct} reflects interfacial polarization, indicating less efficient charge transfer at the catalyst-electrolyte interface. Such changes may arise from catalyst poisoning, deactivation, or mass transport limitations. The observed small increase in R_s and large increase in R_{ct} post-electrolysis suggest that performance loss is dominated by degradation processes localized at the catalytic interface, potentially involving species adsorption or catalyst structural changes. The positively charged amino functional groups can orient and electromigrate toward the cathode, accumulating at the catalyst-electrolyte interface and reducing catalytically active sites. Larger amino acids increase solution viscosity,[31] which hinders CO_2 mass transport and reduces ionic conductivity.[32] Furthermore, amino acids with higher amino-group acid dissociation constant (pK_a) values are more nucleophilic and can reduce local CO_2 availability,[33] while those with lower pK_a values can enhance hydrogen evolution by direct proton donation.[34] Although these effects do not explain long-term degradation, they informed the solvent selection: GLY was chosen for its small size and moderate pK_a, properties that can reduce its initial negative impact on the catalyst-electrolyte interface and attain a higher baseline CO selectivity.

We hypothesized that reducing the GLY concentration would mitigate its negative effects, primarily by limiting its adsorption coverage on the active surface. Maintaining 2.0 M KOH in the capture solvent, we varied the GLY concentration from 0.1 to 3.0 M to examine its effect on electrolysis stability (Figure 2C). Despite containing the least amino acid, the 0.1 M GLY exhibited the shortest operational stability—only 3 h. In contrast, the 3.0 M GLY system attained a CO FE of 45%, which started declining after 25 h.

The primary degradation pathway of phthalocyanine-based catalysts is via demetallation.[35,36] This process is pH dependent, as protons can directly leach the center metal ion and hydroxides can cleave the macrocycle,[37–39] and is accelerated by high cathodic potentials.[40] Inductively coupled plasma-optical emission spectroscopy (ICP-OES) detected dissolved cobalt in the electrolyte after 50 h of operation (Table S3). We therefore reasoned that the instability reflects a limitation intrinsic to RCC broadly: the acidic cation exchange membrane, positioned adjacent to the cathode, delivers protons that regenerate CO_2 near the catalyst interface, shortening the reactant diffusion path but also accelerating cathode degradation. Amino acids, such as GLY, are effective pH buffers due to their functional groups.[41] Therefore, a higher GLY concentration buffered the local pH and slowed proton diffusion to the catalyst surface, leading to improved stability. However, enhanced buffering also introduces trade-offs. The strongly buffered system exhibited a lower peak CO FE, most likely due to less acidic conditions that reduced the supply of *in-situ*-regenerated CO_2 while increasing solution viscosity that hindered CO_2 mass transport to the catalyst. Additionally, high buffering capacity may slow CO_2 capture kinetics, leading to a gradual increase in the initial CO selectivity as the system approaches equilibrium CO_2 loading. Together, these competing effects resulted in improved stability but reduced peak CO FE, highlighting that the buffer concentration should be optimized to balance both performance metrics. Studying phosphate and citrate as alternative buffers in 2 M K-GLY, we observed similar stability trends (Figure S11): increasing the solution buffering capacity improved the stability from 13 to ~30 h. These results suggest that, for small amino acids with moderate pK_a values, such as GLY, local pH, rather than interactions at the catalyst-electrolyte interface, dominates the degradation mechanism. We further selected tripotassium phosphate (K_3PO_4) as the supporting electrolyte due to its strong buffering capacity and small size, then tuned the concentrations of GLY and K_3PO_4 for high CO selectivity and durability (Figures S12–S14).

An optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.5 M K_3PO_4 resulted in 46 h of stable CO production with ~50% CO FE at 100 mA cm^{-2} (Figure 2D). The optimized solvent showed smaller R_s and R_{ct} than 2 M K-GLY, when comparing

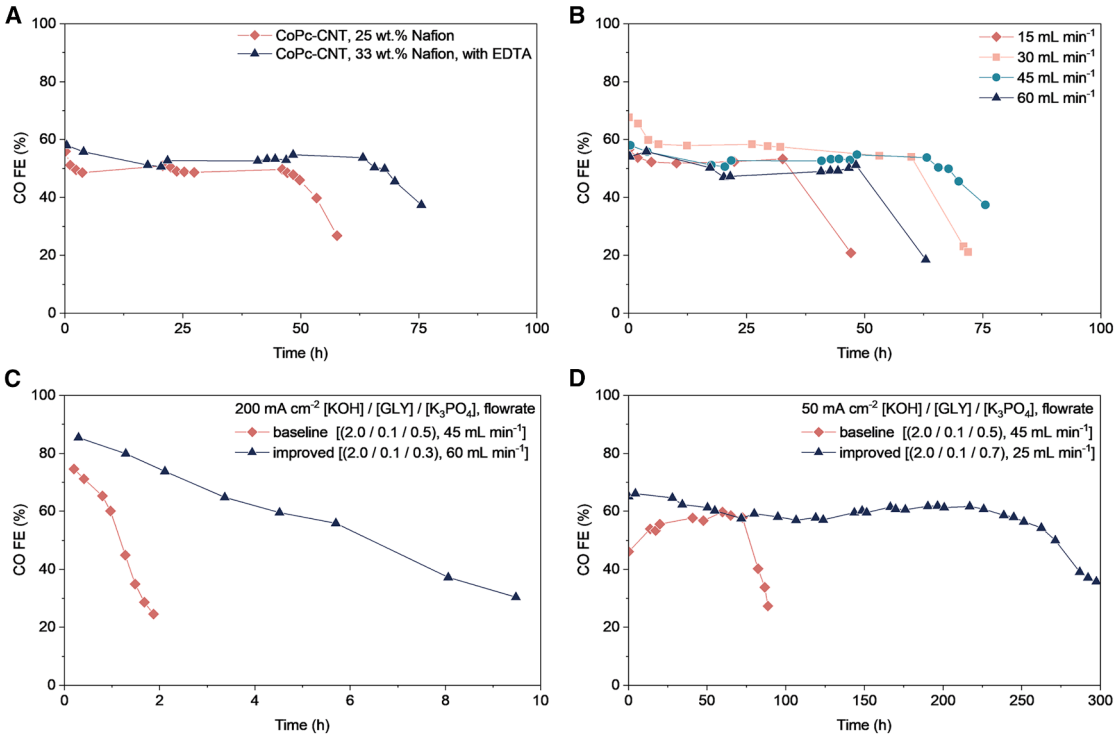

Figure 3. Extending stable CO production under different operating conditions

All experiments were conducted using a 1 cm^2 active area.

(A) CO FE over time using a baseline CoPc-CNT catalyst and one with increased catalyst confinement via added Nafion and ethylenediaminetetraacetic acid (EDTA).

(B) CO FE over time with catholyte flow rates ranging from 15 to 60 mL min^{-1}.

(C) CO FE over time at 200 mA cm^{-2} using the capture solvent and flow rate tailored for 100 mA cm^{-2} (baseline) and an optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.3 M K_3PO_4 operated at 60 mL min^{-1} (improved).

(D) CO FE over time at 50 mA cm^{-2} using the capture solvent and flow rate tailored for 100 mA cm^{-2} (baseline) and an optimized capture solvent consisting of 2.0 M KOH, 0.1 M GLY, and 0.7 M K_3PO_4 operated at 25 mL min^{-1} (improved).

their pre- and post-electrolysis EIS spectra (Figure S15). The addition of phosphate improved solution conductivity and sustained catalyst activity over time. We modeled the pH within the electrolyzer and found that the optimized solvent increased the cathode pH from 11.0 to 11.6 while expanding the alkaline region toward the cation exchange membrane (Figures S16 and S17; Table S3; Note S1). The acidic pH range of 1.6–3.0 allows both GLY (pK_a 2.34) and phosphate (pK_a 2.1) to buffer effectively, as their pK_a values fall within this range. The same solvent also enhanced CO selectivity when an Ag catalyst was used, demonstrating the broad utility of solvent tuning (Figure S18). Nonetheless, strong Ag-amine interactions triggered a rapid decline in CO FE and large R_s and R_{ct} (Figure S19), suggesting that Ag may be incompatible with amine-based solvents.[42]

Catalyst confinement

Even still, after ~50 h of electrolysis using the CoPc-CNT catalyst, we found that dark deposits accumulated in the electrolyte flow channel and metallic clusters in the catalyst (Figures S20 and S21), most likely as a result of continuous exposure to the alkaline electrolyte.[43,44] To reinforce the catalyst layer, we tuned the Nafion ionomer content to enhance catalyst structural integrity (Figure S22).[45–47] Increasing the Nafion content from 25 to 35 wt % enhanced stability, but further increases led to reduced CO FE and diminished stability. Nafion is an excellent ionic conductor, and excess Nafion can increase cathode proton flux, accelerating catalyst degradation pathways. Furthermore, its hydrophobic fluorinated backbone can hinder mass transport of chemisorbed CO_2, resulting in decreased *in situ* CO_2 regeneration.[48] We further incorporated ethylenediaminetetraacetic acid (EDTA) into the catalyst as a chelating agent to limit metal-cluster growth (Figure S23).[49] Although the solubility and mobility of EDTA may increase at higher pH,[50] its strong chelation capacity enables sequestration of dissolved metal species and prevents their deposition onto the catalyst surface, and it has been reported to enhance catalyst durability.[51,52]

This confinement strategy extended stability to 67 h at ~50% CO FE (Figure 3A). Compared to the baseline CoPc-CNT catalyst, the confined catalyst exhibited larger initial resistances, most likely due to the added Nafion and EDTA, but the R_{ct} was smaller than that of the baseline after the same operation time (Figure S24). ICP-OES showed a 4-fold lower concentration of dissolved cobalt in the electrolyte with the confined catalyst (Table S4). Other strategies to improve catalyst stability include

minimizing contaminations in the synthesis process[53] and structural tuning of the catalyst via ligand functionalization and heat treatment.[54,55]

Optimization of operating parameters

Tuning operating parameters further improved durability. An intermediate catholyte flow of 45 mL min^{-1} offered the longest stability (Figure 3B). The electrolyte flow rate directly impacts the CO_2 mass transfer and local pH. A higher flow rate increases the concentration of dissolved CO_2 at the cathode,[56] but it also reduces the cathode alkalinity by refreshing the electrolyte and introduces greater mechanical turbulence. Conversely, a slower flow rate promotes the accumulation of hydroxides, leading to a more alkaline environment.[57] Either extreme shortens the stability. Tuning the flow rate can balance CO_2 availability and local pH,[15] thereby optimizing overall system performance and stability. Potential pulsing, effective in some gas-phase CO_2 electrolyzers,[58] proved detrimental in this system. Cycling between 100 mA cm^{-2} and open circuit (0 mA cm^{-2}) deteriorated the stability, most likely due to halted hydroxide generation during the off state, creating a transiently more acidic surface that accelerated catalyst degradation (Figure S25).

We then investigated the stability at 200 and 50 mA cm^{-2}. The hydroxide generation rate, and thereby the local pH, is governed by the current density. With a more alkaline cathode at 200 mA cm^{-2},[59] a less buffered capture solvent and a faster flow rate were required than for 100 mA cm^{-2} (Figures 3C and S26). Similarly, the stability at 50 mA cm^{-2} was extended to 250 h with CO FE ~60% (CO partial current density of ~30 mA cm^{-2}), with higher buffering capacity and a slower flow rate compared to the baseline (Figures 3D and S27). This stability marked an 8-fold improvement over the previous best AAS-based RCC system. The lower cell voltage at 50 mA cm^{-2} likely also contributed to the slowed CoPc-CNT degradation, consistent with the observed correlation between rising overpotential and declining CO selectivity.

Closed-loop capture and conversion

In addition to achieving year-scale durability on par with commercial water electrolyzers,[60] industrialization of RCC requires integration with existing energy infrastructure. Upstream, a high CO_2 capture efficiency is needed to minimize absorber capital costs,[61] which requires the post-electrolysis solution to be sufficiently depleted of CO_2. Impurities in the industrial CO_2 stream, such as oxygen, nitrogen oxides, and sulfur oxides, may decrease the selectivity of gas-phase CO_2 conversion.[62–64] However, their effects are greatly reduced in liquid-fed RCC systems and can be further managed through system and catalyst design.[65–67] Downstream, high electrolyzer CO_2 utilization ensures that the product stream is CO_2 lean and can integrate with chemical upgrading processes without further purification. These upgraders typically operate at higher pressures and temperatures,[68] requiring RCC electrolyzers to also be stable under elevated conditions to enable direct integration.

To simulate an integrated, closed-loop process, we continuously circulated the catholyte between a CO_2 absorber, an electrolyzer, and a gas-liquid separator (Figure 4A). The CO_2 absorber was sparged with 15 vol % CO_2 at 20 sccm to maintain CO_2 equilibrium, and the carbon efficiencies were evaluated at steady state. The CO_2 capture efficiency was calculated using the difference between CO_2 flowing into and out of the absorber (Equation 1). The CO_2 utilization of the electrolyzer is indicative of the product's gas purity and was calculated using the ratio between the converted and the total CO_2 exiting the gas-liquid separator (Equation 2). The overall CO_2 utilization is the product of CO_2 capture efficiency and electrolyzer CO_2 utilization, representing the carbon efficiency of the integrated closed-loop RCC system (Equation 3).

$$CO_2 \text{ capture efficiency } (\%) = \frac{[CO_2 \text{ IN}] - [CO_2 \text{ OUT}]}{[CO_2 \text{ IN}]} \times 100\% \quad \text{(Equation 1)}$$

$$\text{Electrolyzer } CO_2 \text{ utilization } (\%) = \frac{[CO_2 \text{ converted}]}{[CO_2 \text{ converted}]+[CO_2 \text{ unconverted}]} \times 100\% \quad \text{(Equation 2)}$$

$$\text{Overall } CO_2 \text{ utilization } (\%) = CO_2 \text{ capture efficiency} \times \text{electrolyzer } CO_2 \text{ utilization} \quad \text{(Equation 3)}$$

Employing the optimized capture solvent and confined catalyst with a 1 cm^2 electrolyzer at 100 mA cm^{-2}, we observed a 91% electrolyzer CO_2 utilization, producing a syngas stream containing ~5 vol % CO_2—below the <10 vol % threshold for direct downstream Fischer-Tropsch or methanol synthesis.[69,70] This highly concentrated syngas eliminates the need for CO_2 removal, which is the most energy-intensive downstream separation step in gas-phase CO_2 electrolyzers.[4] The integrated capture and conversion system simplifies upstream and downstream processes to achieve energy-efficient CO production (Figure S28; Table S5; Note S2). However, the CO_2 capture efficiency was 28% (Figure 4B), well below the industrial absorber target of >85% CO_2 removal,[71] resulting in an overall CO_2 utilization of 25%. The capture efficiency can be increased by improving the CO_2 absorption column design and generating more CO_2-lean capture solvents at the RCC electrolyzer outlet.[72,73]

As the capture solvent becomes more saturated with chemisorbed CO_2, its pH and capacity plateaus (Figure S29), demanding higher CO_2 conversion rates to sufficiently regenerate the solvent.[74] Scaling to an active area of 5 cm^2 raised the CO_2 conversion rate, delivering a CO_2 utilization of 92% and an improved CO_2 capture efficiency of 38%, corresponding to an overall CO_2 utilization of 35% (Figure S30; Table S6). A GLY concentration of 0.1 M is ~5 wt %, on par with industrial standards for blended solvents.[75] Though K-GLY remains an effective CO_2 capture activator even at low concentrations,[73] fully integrating RCC with upstream capture will require further solvent engineering and greater CO output—via larger electrodes, higher current densities, and higher CO selectivity.

In the full-cycle configuration, we sustained over 75 h of continuous operation with ~60% CO FE at 1 cm^2—~6 times longer than the initial demonstration in 2 M K-GLY (Figure 4C). At 5 cm^2, although the initial CO FE reached >75%, it started

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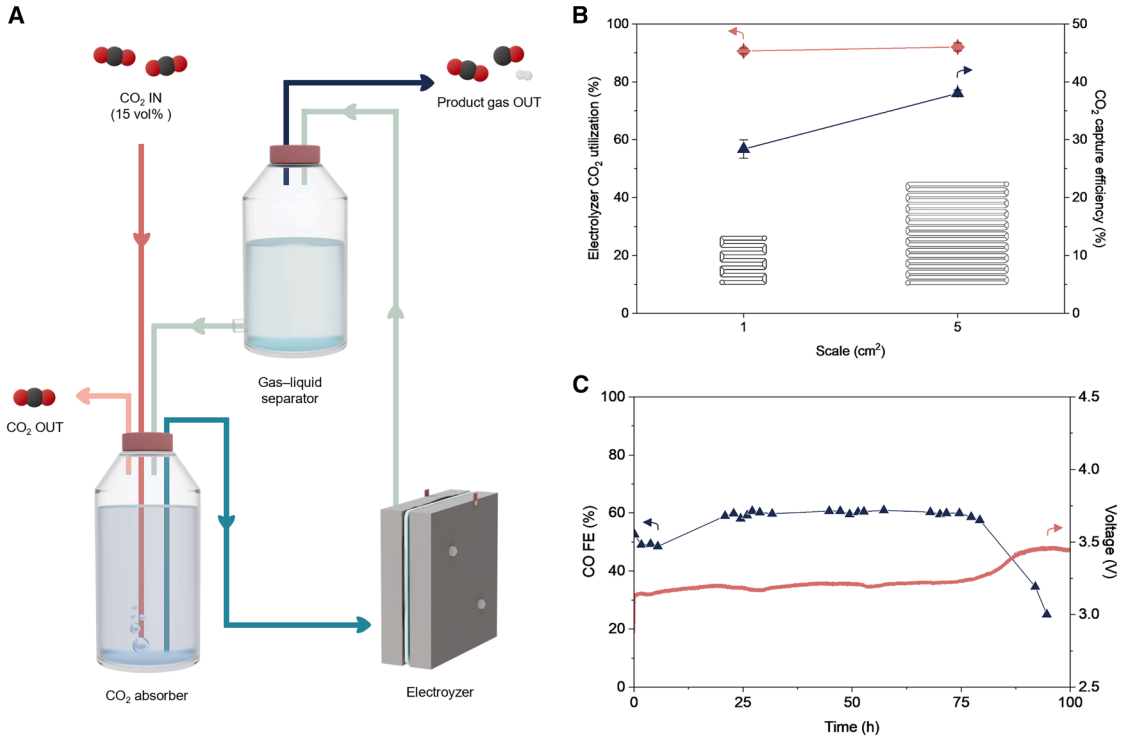

Figure 4. Full-cycle configuration demonstration

All experiments were conducted at an applied current density of 100 mA cm^{-2}, using the optimized capture solvent and confined catalyst.

(A) Schematics of the closed-loop full-cycle system that continuously circulates the catholyte between a CO_2 absorber, an electrolyzer, and a gas-liquid separator. The gas stream "product gas OUT" was quantified to calculate the CO_2 utilization. The gas streams "CO_2 IN" and "CO_2 OUT" were quantified to calculate the CO_2 capture efficiency.

(B) CO_2 utilization and capture efficiency when using a 1 and 5 cm^2 electrolyzer active area. Error bars correspond to the standard deviation of at least three independent measurements.

(C) CO FE and non-iR-compensated full-cell operating voltage over time in a 1 cm^2 electrolyzer.

degrading after 1 h of electrolysis (Figure S31). Achieving stable CO production at larger scales will require further optimization to address scale-dependent challenges: longer flow paths increase concentration gradients[76] and uneven catalyst distribution and local current hotspots can destabilize performance.[77]

Conclusions

The RCC architecture presents an intrinsic trade-off: the protons needed for *in situ* CO_2 regeneration at the cathode simultaneously accelerate catalyst degradation. In prior gas-phase CO_2 electrolyzers with acidic cathodes, stability has been improved through catalyst engineering, such as interface modification with an immobilized cationic layer to increase the local pH and the design of an acid-stable precursor that reduces *in situ* to generate a high density of durable active sites.[78,79] The capture solvent tuning strategy presented in this study is a system-level approach for RCC that complements material advances by offering a distinct set of tunable parameters and enables optimization of the integrated capture and conversion processes.

We showed that fine control of the cathode microenvironment—particularly local pH—is key for durable RCC. By formulating an amine-based capture solvent that balances buffering strength while minimizing interactions at the catalyst-electrolyte interface and by co-optimizing flow rate with current density to match hydroxide generation, we sustained CO formation far beyond previous benchmarks (Figure S1; Table S1). These advances translated into stable operation for over 75 h at 100 mA cm^{-2} with a CO partial current density of ~60 mA cm^{-2}—~6 times longer than initial results—and over 250 h at 50 mA cm^{-2} with a CO partial current density of ~30 mA cm^{-2}. The optimized capture solvent regulated proton flux by employing buffers with pK_a values that matched the local acidic pH. This pH-tuning strategy can be extended to a broader class of electrolyzers that need to balance sharp pH gradients for efficiency against the pH-sensitive stability of their components. In a closed absorber-electrolyzer system, the CO_2 capture and utilization efficiencies were improved by increasing the total CO_2 conversion rate via a larger electrolyzer active area. At 5 cm^2, we achieved a capture efficiency of 38% and an electrolyzer CO_2 utilization of 92%, resulting in an overall CO_2 utilization of 35%. Identifying and understanding the system parameters that affect stability and carbon efficiency can guide the design of more robust RCC systems for commercialization.

While the results from this study are promising, further development is required in several key areas. The first is designing an

intrinsically stable catalyst. Though the confinement strategy extended the durability of the CoPc-CNT catalyst, the CO FE decreased more aggressively at higher current densities. The ideal catalyst is chemically tolerant of the capture solvent, impurities, and operating pH and mechanically robust under the operating voltage, flow pressure, and gas bubble nucleation. The second is maintaining stable operation at scale. When the optimized system was directly implemented with a 5-fold larger active area, performance degraded rapidly. Scaling introduced non-linear, fundamental changes to the electrochemical environment, such as shifts in current distribution, mass transport, or local pH, and thus needs to be addressed through dedicated scaling strategies. Lastly, for the economic integration of CO_2 capture and conversion, the RCC electrolyzer must regenerate the capture solvent to a sufficiently CO_2-lean state to enable efficient upstream capture. The demonstrated CO_2 capture efficiency currently falls below industrial targets but can be increased through further solvent optimization and by achieving higher CO_2 conversion rates in the electrolyzer. Addressing these interconnected challenges in stability, scale, and integration through materials and systems innovations will advance the industrial viability of RCC.

METHODS

Materials

Unless otherwise specified, all reagents were purchased from commercial suppliers without further purification. Milli-Q deionized (DI) water with a resistivity of 18.2 MΩ at 25°C was used for all experiments.

CoPc-CNT catalyst preparation

Catalysts were synthesized with modification from previous reports.[80,81] 5 mg of CoPc (Thermo Fisher Scientific) was dispersed in 20 mL of N,N-dimethylformamide (DMF; Sigma-Aldrich) and sonicated for 1 h. 30 mg of carboxylic-acid-functionalized multi-walled CNT (Sigma-Aldrich) was dispersed in 20 mL of DMF and sonicated for 1 h. The CoPc-containing solution was added dropwise into the CNT-containing solution and sonicated for 1 h. After sonication, the mixture solution was stirred for 24 h at 600 rpm at room temperature. The CoPc-CNT was separated by vacuum filtration, washed with DMF, ethanol, and DI water, and air-dried in an oven at 60°C overnight.

CoPc-CNT catalyst characterization

High-resolution transmission electron microscopy (TEM) and high-angle annular dark-field scanning TEM (HAADF-STEM) images were acquired at 300 kV using a Thermo Fisher Scientific Spectra Ultra microscope equipped with a probe spherical-aberration corrector and a high-brightness, ultra-stable monochromated Schottky field-emitter gun.

Cathode fabrication

The Ag catalyst ink was prepared using 25 mg of Ag nanoparticles (Ag NPs; 20 nm, US Research Nanomaterials), 75 mg of Nafion dispersion (D521, 5 wt %, 1100 EW, Ion Power), and 3 mL of ethanol for a 6.25 cm^2 substrate. The Ag catalyst ink was sonicated for 1 h prior to airbrushing onto carbon paper (AvCarb MGL 190, Fuel Cell Store) to obtain a catalyst loading of ~2 mg cm^{-2}. The baseline CoPc-CNT catalyst ink was prepared using 5 mg of the synthesized CoPc-CNT, 33 mg of Nafion dispersion, and 3 mL of ethanol for a 6.25 cm^2 substrate. The baseline CoPc-CNT catalyst ink was sonicated until fully dispersed (~6 h) prior to airbrushing onto carbon paper to obtain a catalyst loading of ~1 mg cm^{-2}. The confined CoPc-CNT catalyst ink was first prepared using 5 mg of the synthesized CoPc-CNT, 50 mg of Nafion dispersion, and 3 mL of ethanol for a 6.25 cm^2 substrate and sonicated until fully dispersed (~6 h). Then, 5 mg of ethylenediaminetetraacetic acid disodium salt dihydrate (Na_2EDTA; BioShop) was added to the dispersed CoPc-CNT catalyst ink and sonicated until the Na_2EDTA fully dissolved (~24 h).[49] The confined CoPc-CNT catalyst was airbrushed onto carbon paper to obtain a catalyst loading of ~1.3 mg cm^{-2}. All airbrushed cathodes were air-dried in an oven at 60°C overnight.

Electrolyzer operation

A membrane electrode assembly was used for all electrolysis experiments. The fabricated cathode and titanium-supported iridium oxide anodes (Magneto Special Anodes, Evoqua Water Technologies) had an exposed area of either 1 or 5 cm^2 and were separated by a Nafion membrane (N117, Ion Power). A mixed cellulose ester membrane filter (8.0 μm pore size, Millipore Sigma) was added as a carbon dioxide (CO_2) diffusion layer between the cathode and the membrane. Nafion membranes were activated via the following procedure: 1 h at 90°C in 3 wt % hydrogen peroxide (H_2O_2), 1 h at 90°C in DI water, 1 h at 90°C in 5 wt % sulfuric acid (H_2SO_4, 95%), and 1 h at 90°C in DI water again. The activated Nafion membranes were stored in DI water.

In a typical experiment, the catholyte was prepared by first purging 85 mL of the capture solvent with 100% CO_2 at 200 sccm for ~2 h. Then, the CO_2-rich post-capture solution was purged with 20 sccm of 15 vol % CO_2 (17 sccm nitrogen [N_2] and 3 sccm CO_2) until the pH of the solution reached a plateau (~1 h). The pH of the solution was measured using a pH/conductivity meter with an accuracy of ±0.01 (PC8500, Apera Instruments). The AAS-based solvents were prepared using KOH (>85%, Sigma-Aldrich), GLY (>99.5%, BioShop), LYS (>98%, Sigma-Aldrich), PRO (>98.5%, BioShop), SER (>98.0%, BioShop), SAR (>98%, Sigma-Aldrich), dipotassium phosphate (K_2HPO_4, >98%, BioShop), and tripotassium citrate ($K_3C_6H_5O_7$, reagent grade, BioShop) in various combinations and ratios. The pure (bi)carbonate experiments used potassium carbonate (K_2CO_3, ACS grade, BioShop) or potassium bicarbonate ($KHCO_3$, ACS grade, BioShop) salts as the catholyte. The anolyte was 0.05 M H_2SO_4. Both catholyte and anolyte were controlled by peristaltic pumps. During electrolysis, the catholyte was continuously fed with 20 sccm of 15 vol % CO_2 to maintain CO_2 equilibrium.

The electrolysis data were collected using a potentiostat (Autolab PGSTAT204) or a battery tester (LANDT CT3002A) in galvanostatic mode. All reported full-cell operating voltages were not iR corrected. The EIS analysis was conducted with the potentiostat, acquired in the frequency range of 10^5–1 Hz at 100 mA cm^{-2}, with a geometric active area of 1 cm^2. A Randles circuit with a constant phase element was used for circuit fitting and

to obtain R_s and R_{ct}. The first gas samples were collected after 15 min of operation to ensure even mixing of gaseous products. All experiments were conducted at ambient temperatures and pressures.

Gas and liquid analysis

Gas chromatography (GC; PerkinElmerClarus 590) coupled with a thermos conductivity detector (TCD) and flame ionization detector (FID) was used to quantify the gas products, carbon monoxide (CO), hydrogen (H_2), and methane (CH_4), as well as CO_2 in the gas streams. Gas samples were analyzed in 1 mL volume, and the FEs toward gas products were calculated using the following Equation 4:

$$FE_i\,(\%) = \frac{z_i F P}{RT} \times \nu_i \times \frac{1}{I} \times 100\%\,, \qquad \text{(Equation 4)}$$

where z_i represents moles of electrons needed to produce one mole of the product i, F is the Faraday constant (96,485 C mol^{-1}), P is the pressure (101.325 kPa), R is the ideal gas constant (8.314 J mol^{-1} K^{-1}), T is the temperature (293 K), ν_i is the gas flow rate of the product i, and I is the total current. The gas flow rate was measured using a bubble column. Only trace amounts of CH_4 (<0.1%) and no other gaseous CO_2 electrolysis products were detected.

Proton nuclear magnetic resonance (^{1}H-NMR) in water-suppression mode was used to quantify possible liquid electrolysis products (e.g., formate) and identify the chemical species before and after CO_2 capture. Carbon NMR (^{13}C-NMR) was used to identify the chemical species before and after CO_2 capture and to quantify the amount of carbamate and bicarbonate in the post-capture solution. NMR was performed on an Agilent DD2 600 spectrometer in deuterium oxide (D_2O), with trimethylsilylpropanoic acid (TSP) as the internal standard. No liquid CO_2 electrolysis products were detected.

ICP-OES (Thermo Scientific iCAP Pro) was used to quantify the concentration of cobalt in the electrolyte before and after prolonged electrolysis. 0.5 mL of the electrolyte was mixed in a Teflon digestion vessel with 2 mL of nitric acid (HNO_3, 65%) and 1 mL of hydrochloric acid (HCl, 37%). The mixture was heated at 120°C for 2 h to completely decompose any organic components. After cooling, the samples were diluted to 50 mL with DI water and filtered through a polytetrafluoroethylene membrane (0.22 μm pore size, Millipore Sigma).

Full-cycle configuration

The catholyte was prepared by first purging 1 L of the optimized capture solvent, consisting of 2.0 M KOH, 0.1 M GLY, and 0.3 M K_3PO_4, with 100% CO_2 at 200 sccm for ~8 h. The CO_2-rich post-capture solution was saturated with 20 sccm of 15 vol % CO_2 until the pH of the solution reached a plateau. The solution was then split between the CO_2 absorber and the gas-liquid separator. During electrolysis, the catholyte was pumped from the CO_2 absorber, through the cathode of the electrolyzer, to the gas-liquid separator. The gas-liquid separator was placed higher than the CO_2 absorber to allow the catholyte to flow back into the CO_2 absorber by gravity. The addition of a gas-liquid separator downstream of the electrolyzer enabled the quantification of CO_2 utilization by the electrolyzer and CO_2 capture efficiency by the absorber. Gas samples were taken from the outlet of the gas-liquid separator and quantified using GC to calculate the CO_2 utilization. The CO_2 flowing into and out of the absorber was quantified using GC to calculate the CO_2 capture efficiency.

RESOURCE AVAILABILITY

Lead contact

Further information and request for resources should be directed to and will be fulfilled by the lead contact, David Sinton (dave.sinton@utoronto.ca).

Materials availability

All reagents in this study are either commercially available or can be prepared as indicated in the methods section.

Data and code availability

The data supporting the findings of this study are available in the article and supplemental information or from the corresponding authors upon reasonable request.

ACKNOWLEDGMENTS

The authors gratefully acknowledge support from Shell Global Solutions International B.V., the Canada Research Chairs Program, and the Natural Sciences and Engineering Research Council of Canada through the Alliance Program and the Discovery Grant Program. Y.C.X. thanks NSERC and Hatch for their support through graduate scholarships.

AUTHOR CONTRIBUTIONS

Conceptualization, Y.C.X., R.K.M., and D.S.; methodology, Y.C.X., C.M.G., K.H., P.J.C., and R.K.M.; investigation, Z.G., S.S.S., M.L., H.S.L., J.L., C.W., D.M.V., and Y.Z.; computational modeling, R.W. and F.L.; visualization, Y.C.X.; funding acquisition, D.S.; supervision, R.K.M. and D.S.; writing – original draft, Y.C.X., R.K.M., and D.S.; writing – review & editing, all authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.checat.2026.101694.

Received: October 15, 2025
Revised: December 4, 2025
Accepted: March 5, 2026

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